Economics and Business Administration — A.Y. 2025-26

Mock Exam — Solutions

Exercise 1)

The reclassified balance sheet of Società Alfa is as follows:

BALANCE SHEET				
	Year X-1	**Year X-1 %**	**Year X**	**Year X %**
ASSETS				
Cash and cash equivalents	225	*0.1%*	0	*0.0%*
Trade receivables	35,046	*20.1%*	68,750	*27.6%*
Inventories	25,225	*14.4%*	72,880	*29.2%*
Other current assets	450	*0.3%*	200	*0.1%*
CURRENT OPERATING ASSETS	**60,946**	***34.9%***	**141,830**	***56.8%***
Property, Plant & Equipment	108,187	*61.9%*	101,725	*40.8%*
Intangible assets	0	*0.0%*	0	*0.0%*
NON-CURRENT OPERATING ASSETS	**108,187**	***61.9%***	**101,725**	***40.8%***
NON-CORE ASSETS	**5,626**	***3.2%***	**5,988**	***2.4%***
TOTAL ASSETS	**174,759**	***100.0%***	**249,543**	***100.0%***
LIABILITIES & EQUITY				
Trade payables (goods and services)	30,926	*17.7%*	41,621	*16.7%*
Tax payables, other & TFR	28,947	*16.6%*	28,096	*11.3%*
CURRENT OPERATING LIABILITIES	**59,873**	***34.3%***	**69,717**	***27.9%***
Short-term bank debt	60,163	*34.4%*	127,850	*51.2%*
Medium/long-term bank debt	17,000	*9.7%*	11,592	*4.6%*
Other financial liabilities	978	*0.6%*	978	*0.4%*
FINANCIAL LIABILITIES (D)	**78,141**	***44.7%***	**140,420**	***56.3%***
Share capital & reserves	39,025	*22.3%*	36,745	*14.7%*
Net income for the period	-2,280	*-1.3%*	2,661	*1.1%*
EQUITY (E)	**36,745**	***21.0%***	**39,406**	***15.8%***
TOTAL LIABILITIES & EQUITY	**174,759**	***100.0%***	**249,543**	***100.0%***

INCOME STATEMENT				
	Year X-1	**Year X-1 %**	**Year X**	**Year X %**
Net Sales	**218,125**	***100.00%***	**335,550**	***100.00%***
Purchases of goods	-100,000	*-45.85%*	-150,000	*-44.70%*
Purchases of services	-53,802	*-24.67%*	-85,640	*-25.52%*
Change in inventories	-5,995	*-2.74%*	-7,985	*-2.38%*
Labour costs	-48,038	*-22.02%*	-66,740	*-19.89%*
Depreciation, amortization & write-downs	-10,412	*-4.77%*	-18,950	*-5.65%*
Core EBIT (EBIT_core)	**-122**	***-0.06%***	**6,235**	***1.86%***
Income from ancillary activities	3,110	*1.43%*	3,813	*1.14%*
EBIT (Operating Income)	**2,988**	***1.37%***	**10,048**	***2.99%***
Net Financial Expenses	-3,980	*-1.82%*	-6,675	*-1.99%*
EBT (Earnings Before Taxes)	**-992**	***-0.45%***	**3,373**	***1.01%***
Income Taxes	-1,288	*-0.59%*	-713	*-0.21%*
NET INCOME (NI)	**-2,280**	***-1.05%***	**2,661**	***0.79%***

Question 1

a) Calculate the following financial ratios:

	Year X-1	**Year X**
Operating NWC	60,946 − 59,873 = 1,073	141,830 − 69,717 = 72,113
DSO (incl. VAT 20%) = (Trade receivables / 1.20) / (Net Sales / 360)	(35,046/1.20) / (218,125/360) = 48.2 days	(68,750/1.20) / (335,550/360) = 61.5 days
DPO (incl. VAT 20%) = (Trade payables / 1.20) / ((Purchases goods + services) / 360)	(30,926/1.20) / (153,802/360) = 60.3 days	(41,621/1.20) / (235,640/360) = 53.0 days
DIO = Inventories / (Cost of goods sold* / 360)	25,225 / (105,995/360) = 85.7 days	72,880 / (157,985/360) = 166.1 days
Working Capital Cycle (DSO − DPO + DIO)	48.2 − 60.3 + 85.7 = 73.6 days	61.5 − 53.0 + 166.1 = 174.6 days

** Cost of goods sold = Purchases of goods + Change in inventories*

b) Based on the ratios calculated and the financial statement data (note in particular the trend in net sales), comment on the causes behind the increase in Operating NWC. (max. 5 lines)

Against a significant increase in net sales (+54%), the period shows an increase in DSO and a reduction in DPO, alongside a near-doubling of DIO. The build-up of inventories appears to be the primary driver of the increase in the Working Capital Cycle and in the overall Operating NWC balance.

Question 2

a) Calculate ROE and decompose it into its drivers using the financial leverage formula (Modigliani-Miller)

	Year X-1	**Year X**
ROE = NI / Equity (E)	-2,280 / 36,745 = −6.2%	2,661 / 39,406 = 6.8%
CE = Total Assets − Current Operating Liabilities	174,759 − 59,873 = 114,886	249,543 − 69,717 = 179,826
RONA = EBIT / CE	2,988 / 114,886 = 2.60%	10,048 / 179,826 = 5.59%
i = Net Financial Expenses / D	3,980 / 78,141 = 5.09%	6,675 / 140,420 = 4.75%
Leverage ratio (D/E)	78,141 / 36,745 = 2.12×	140,420 / 39,406 = 3.56×
Tax Incidence (NI / EBT)	−2,280 / −992 = 229.8%	2,661 / 3,373 = 78.9%
FORMULA DECOMPOSITION ROE = [RONA + (RONA−i) × D/E] × NI/EBT	**(2.60%+(2.60%−5.09%)×2.12) × 229.8% = −6.2%**	**(5.59%+(5.59%−4.75%)×3.56) × 78.9% = 6.8%**

b) Based on the ratios calculated, comment on the "levers" used by the company to increase ROE in Year X. (max. 5 lines)

The shift from a negative to a positive ROE in Year X is driven by the return to profitability, and significantly by the increase in financial leverage (D/E ratio: 2.12× → 3.56×). Since RONA > i in Year X (positive spread of +0.84 pp), the greater use of debt amplifies ROE above RONA: financial leverage is working in shareholders' favour.

Exercise 2)

The reclassified balance sheet of Società Commerciale Beta (Capital Employed approach) is as follows:

BALANCE SHEET		
	Year X-1	**Year X**
ASSETS		
Cash and cash equivalents	54	73
Trade receivables	4,003,082	8,860,000
Inventories	1,840,000	3,435,000
CURRENT OPERATING ASSETS	**5,843,136**	**12,295,073**
Property, Plant & Equipment	6,245,000	9,843,000
Intangible assets	24,500	36,000
NON-CURRENT OPERATING ASSETS	**6,269,500**	**9,879,000**
NON-Core ASSETS	**3,800,230**	**127,000**
TOTAL ASSETS	**15,912,866**	**22,301,073**
LIABILITIES & EQUITY		
Trade payables (goods and services)	4,112,236	5,629,150
Tax payables, other & TFR	205,830	214,775
CURRENT OPERATING LIABILITIES	**4,318,066**	**5,843,925**
Short-term bank debt	2,154,800	7,540,420
Medium/long-term bank debt	4,725,000	4,004,728
FINANCIAL LIABILITIES (D)	**6,879,800**	**11,545,148**
Share capital & reserves	4,500,000	4,500,000
Net income for the period	215,000	412,000
EQUITY (E)	**4,715,000**	**4,912,000**
TOTAL LIABILITIES & EQUITY	**15,912,866**	**22,301,073**

INCOME STATEMENT		
	Year X-1	**Year X**
Net Sales	**23,789,900**	**29,758,850**
Cost of goods sold	-16,724,000	-19,756,000
Purchases of services	-2,594,000	-4,308,000
Labour costs	-2,065,000	-3,875,312
Depreciation & amortization	-945,800	-1,819,455
Core EBIT (EBIT_core)	**179,728**	**1,738,538**
Income from ancillary activities	342,000	12
EBIT (Operating Income)	**521,728**	**1,738,550**
Net Financial Expenses	-229,443	-964,050
EBT (Earnings Before Taxes)	**292,285**	**774,500**
Income Taxes	-77,285	-362,500
NET INCOME (NI)	**215,000**	**412,000**

** Note: for simplicity, the result of non core activities is represented by the line item "Income from ancillary activities".*

a) Calculate ROI and decompose it into its drivers (core vs. non-core operations):

Question 2a

	Year X-1	**Year X**
ROI = EBIT / CE	521,728 / 11,594,800 = 4.50%	1,738,550 / 16,457,148 = 10.56%
Core ROI = Core EBIT / CE_core CE_core = (Curr. Op. Assets − Curr. Op. Liabilities) + Non-Curr. Op. Assets	179,728 / 7,794,570 = 2.31%	1,738,538 / 16,330,148 = 10.65%
Non-Core ROI = Non-Core EBIT / CE_anc	342,000 / 3,800,230 = 9.00%	12 / 127,000 = 0.01%
Weight of Core Operations (CE_core / CE_tot)	7,794,570 / 11,594,800 = 67.22%	16,330,148 / 16,457,148 = 99.23%
Weight of Non-Core Operations (CE_anc / CE_tot)	3,800,230 / 11,594,800 = 32.78%	127,000 / 16,457,148 = 0.77%
FORMULA DECOMPOSITION ROI = (Core ROI × Weight_core) + (Non-Core ROI × Weight_non-core)	**2.31%×67.22% + 9.00%×32.78% = 4.50%**	**10.65%×99.23% + 0.01%×0.77% = 10.56%**

b) What are the reasons behind the improvement in Core ROI? (max. 5 lines)

The improvement in Core ROI is driven by a better relationship between revenues and operating costs (Core EBIT margin improvement). The company strategically divested most of its ancillary assets and focused resources on core operations, a decision that proved effective given the significant recovery in core operating efficiency.

Solution — Exercise 3

Classify each item using the Capital Employed reclassification scheme, marking with X if the item belongs to the category, or (X) if it is deducted from the category total.

	Operating NWC	Non-Current Operating Assets	Non Core Assets	Financial Liabilities	Core Operating Costs	Equity
Bond investments in portfolio			**X**			
Plant & equipment		**X**				
Surplus cash and cash equivalents			**X**			
Bond loans (issued)				**X**		
Finished goods inventories	**X**					
Current portion of mortgage loans				**X**		
Deferred tax provision	**(X)**					
Advance payments to suppliers for raw materials	**X**					
Legal reserve						**X**
Post-employment benefit obligations (TFR)	**(X)**					
Trade payables	**(X)**					
Transportation costs					**X**	
Trademarks and patents		**X**				